NE585 – Nuclear fuel cycle analysis
Project 6a – Back end of the nuclear fuel cycle

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Total – 350

1 Reactivity & burnup – (100)

Derive an expression for $\rho_n(t)$ in terms of $B(T)$. See **??** for equations.

2 Reactivity swing – (50)

Show that the reactivity swing decreases with increasing batches.

3 Burnup limit – (50)

Find $\lim_{n\to\infty} \frac{B_n(T)}{B(T)}$.

4 Burnup & fuel management – (50)

What does this mean in terms of practical incore fuel management?

5 Small Modular Reactors – (50)

Are there special core loading considerations for NuScale?

6 Burn cycles – (50)

What is the relationship between SWU, extended burnup, and extended burn cycle?

Tables

Figures

Appendix I Acronyms

SMR Small Modular Reactor.